Research Project - Julien Boulanger

My research lies at the crossroads between geometry and dynamical systems. I am particularly interested in surfaces and their interactions with ergodic theory, number theory, and combinatorics. One of my main research topics is the study of translation surfaces and their Veech groups. Informally, a translation surface can be seen as a collection of polygons in the plane with parralel sides idenfied by translations. Such a surface inherits a flat metric except on a finite number of conical singularities whose angles are integer multiples of 2π, and which correspond to certain vertices of the polygons.

During my PhD thesis, I was specifically interested in the algebraic intersection of closed curves on these surfaces, and more precisely the question of knowing *how many times two closed curves of a given length can intersect*. In addition, the second part of my thesis is devoted to the study of *Veech groups* of translation surfaces, which somehow encode the symmetries of each given surface and which are Fuchsian groups. A fundamental (but difficult) question is to know *which Fuchsian groups can be realized as Veech groups of translation surfaces*. During my postdoc at the University of Chile, I continued working on these questions while making new projects. One of these, although motivated by geometry, turns out to be a lot of group theory: the study of automorphisms of a Riemann surface preserving a holomorphic one form. Apart from that, I am studying the algebraic intersection of closed curves in a more general context, for hyperbolic surfaces on the one hand and for surfaces with a Riemannian (resp. Finsler) metric on the other hand. Finally, I am working on several projects related to the ergodic theory of *Interval Exchange Transformations* (IETs).

Organization. In the sequel, I will describe the following.

- In Section A, I focus on the study of translation surfaces and their Veech groups. Namely, I first adress the question of the classification of Teichmüller curves by explaining a result joint with Sam Freedman [BF24] which shows that there is no geometrically primitive Veech surface in the locus $Prym(2, 2) \subset \mathcal{H}(2, 2)$ of surfaces having a *Prym involution*. Then, I adress the question of finding Veech groups which are not finitely generated. This is related to the notion of *connection points* and the distribution of *saddle connections*, and, in the case of surfaces constructed with regular polygons, to the notion of *Hecke continued fractions*. In particular, this exhibits a correspondence with the number theory problem of determining the elements in a given field with a finite Hecke continued fraction expansion, which is an open question that has been studied for more than 70 years now. I developped a geometric approach to this question in [Bou22, Bou25c], and I used this correspondence to exhibit a large family of points which are not connection points while making a precise conjecture (Conjecture 3) on their existence. In this section I describe several questions and ideas related to this conjecture.
- In Section B, I elaborate upon the notion of *interaction strength* which quantifies the maximum number of intersection of two curves of a given length on a Riemannian surface (possibily with conical singularities). Along with several general questions on the interaction strength, I will specifically address the case of translation surfaces and the behavior of the interaction strength by deformation of a given surface, which I studied in [BLM24] (with E. Lanneau and D. Massart), [Bou25b], [Bou24], [BP24] (with I. Pasquinelli) and my recent preprint [Bou25a].
- In Section C, I describe a new preprint [BGRL25] which investigates the *automorphism group* of a translation surface. Since a translation surface can be seen as a pair (X, ω) where X is a Riemann surface and ω a holomorphic one form on X, the automorphism group of the translation surface (X, ω) is a subgroup of $\mathrm{Aut}(X)$.
- Finally, in Section D, I describe a few other projects in different directions, related to *Interval Exchange Transformations* (IETs). In the first project, we aim to study the *Khinchin constant*, coming from an ergodic average with respect to the Zorich measure on IETs. This measure, although intensively studied during the last twenty years, has still a lot to unveil and we propose to explore a (difficult but less investigated) approach using simplices, which could allow to have access to the measure associated with each permutation separately, answering a question of A. Avila. In a second research direction, we aim to study the *abelian complexity* of words coming from interval exchange transformations.

A - Translation surfaces and their Veech groups

This first research direction focuses on the algebraic properties of periodic directions on Veech surfaces. This group can be thought of as the symmetry group associated with a translation surface. More precisely, the (moduli) space of translation surfaces carries a natural action of $GL_2^+(\mathbb{R})$, which comes from the action on the polygons. As shown by Veech, the stabilizer of a given translation surface is a Fuchsian group, which is now referred to as the Veech group of X. Every element of the Veech group induces an[1] *affine diffeomorphism* of the surface. The work of several authors, including Veech, has shown that this group encodes many geometric properties of the surface.

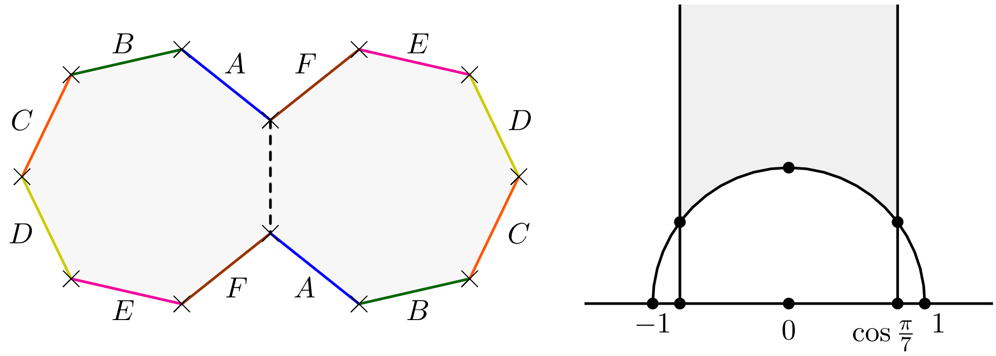

Figure 1: On the left, the double heptagon translation surface. It has genus three and a single singularity of angle 10π. On the right, a fundamental domain in the Poincaré half-plane $\mathbb{H}$ for the orbit of the double heptagon under the action of $SL_2(\mathbb{R})$. Each point corresponds to an element of the orbit and the double heptagon lies at the two identified corners. The Veech group of the double heptagon is conjugated to the Hecke group of level seven. In particular, the double heptagon is a *Veech surface*.

The translation surfaces whose Veech group is a lattice (called Veech surfaces) form a class of particularly interesting surfaces, both from the geometrical and dynamical point of view. The orbit of these surfaces under the action of $SL_2(\mathbb{R}) \subset GL_2^+(\mathbb{R})$ projects in the moduli space $\mathcal{M}_g$ of Riemann surfaces to a complex geodesic, which is called a *Teichmüller curve*. The classification of Teichmüller curves, and more generally of orbit closures under the $SL_2(\mathbb{R})$-action, is a central problem in Teichmüller dynamics where recent progress lead to the developpement of many new and revolutionnary techniques including the work of award winning mathematicians like M. Mirzakhani, C. McMullen and A. Eskin, to name a few.

A.1 - Veech Surfaces with a Prym Involution

One method for classifying Teichmüller curves is to start by searching for those that cannot be constructed from a covering of another Veech surface, which are called *geometrically primitive*. Despite several recent major advances [EFW18] and [EMMW20], this question seems far from being solved.

In genus two, primitive Veech surfaces have been classified in a series of articles by C. McMullen [McM05b], [McM05a], and [McM06a]. In genus three, the classification is not complete, but C. McMullen [McM06b] exhibited an infinite family of primitive Veech surfaces. The work of several people has since shown that, outside of this family, there are only finitely many primitive Veech surfaces (up to the action of $GL_2^+(\mathbb{R})$), see [McM21, Theorem 5.5]. This infinite family consists of surfaces equipped with a *Prym involution* and having a single singularity. A translation surface with a Prym involution can be constructed as a double cover of a *half-translation* surface (branched at the singularities), see [McM06b] or [LN13].

To search for other geometrically primitive Veech surfaces, a natural idea is to focus on translation surfaces equipped with a Prym involution, but with multiple singularities. In this context, E. Lanneau and M. Möller [LM19] studied Veech primitive surfaces in the spaces $Prym(2, 1, 1)$ and $Prym(2, 2)$. They show that there are no Veech primitive surfaces in $Prym(2, 1, 1)$ and identify 92 candidates in $Prym(2, 2)$. In joint work with Sam

[1]In some cases, it may also induce several different affine diffeomorphisms, as we will see in Section 5.

Freedman [BF24], we specifically focus on these 92 candidates and prove that none of them are Veech surfaces, proving:

Theorem 1. *There are no geometrically primitive Veech surfaces in* $Prym(2,2)$*.*

One of the remaining problems in genus three is then:

Question 1. *Are there geometrically primitive Veech surfaces in* $Prym(1,1,1,1)$*?*

In fact, we expect that there are none. Hovewer, this is worth studying since we currently know very few examples of primitive Veech surfaces, and any new example would be remarkable, see the survey [McM21].

A.2 - Periodic Directions and Connection points

In a complementary direction, we can look for translation surfaces whose Veech group is not finitely generated. A way of building such surfaces has been exhibited by P. Hubert and T. Schmidt [HS04], and the construction uses branched covers of Veech surfaces ramified at a (non-periodic) *connection point.* A connection point is a (non-singular) point on a translation surface such that any geodesic originating from a singularity and passing through this point will meet another singularity. In other words, the forward geodesic flow and the backwards geodesic flow from this point (and with any given direction) have the same behaviour. A first motivation to their study is related to billiard trajectories: it is well-known [FK36, ZK76] that billiard trajectories in a *rational polygon*[2] are in correspondence with geodesic trajectories on translation surfaces. In particular, one could be interested in this connection property for rational billards and ask:

Question 2. *What are the connection points on rational triangular billiards? right-angled?*
What are the connection points of a "generic" translation surface?

Most likely, the only connection points of a generic surface are the fixed points of the hyperelliptic involution (when it exists).

Connection points on Veech surfaces. However, if our motivation is the construction of translation surfaces whose Veech group is not finitely generated, the construction of Hubert and Schmidt suggests to focus on the connection points of Veech surfaces. A major result by Veech is that these surfaces have *optimal dynamics*: given a Veech surface X and a direction θ, either all geodesics in direction θ are closed (or are saddle connections, going from one singularity to another, potentially the same), or they are all uniquely ergodic. Moreover, the first case occurs if and only if θ is a proper direction of a parabolic matrix of the Veech group. By identifying the set of directions with $\mathbb{R} \cup \{\infty\} \simeq \partial\mathbb{H}$ via (the inverse of) the slope, the periodic directions of a Veech surface correspond to the *parabolic limit set* of the lattice $SL(X)$. Therefore, the study of connection points on Veech surfaces "reduces" to the study of this set, whether to characterize it, which is a (very) difficult problem, or to deduce geometric properties of translation surfaces from it.

Connection points on regular polygons In fact, there are only two known ways to construct translation surfaces with infinitely generated Veech groups: Hubert-Schmidt's construction and a construction by C. McMullen [McM03] which only work when the so-called *trace field* of the Veech group is quadratic over $\mathbb{Q}$.

In particular, Hubert and Schmidt's construction is supposed to be more general than McMullen's. However, it turns out that we do not know *any* example of non-periodic connection point on a Veech surface whose trace field has degree three or more over $\mathbb{Q}$[3]. My work in [Bou22] and more recently [Bou25c] tackle precisely this question, and we describe in this section a conjecture on the existence of connection points along with several ideas towards the (partial) resolution of this (difficult) conjecture (Conjecture 3).

[2] That is, whose angle are rational multiples of π.
[3] Further, in the quadratic case, connection points are classified as a consequence of the work of Boshernitzan [Bos88], and in this setting Hubert-Schmidt and McMullen's constructions give the same family of surfaces.

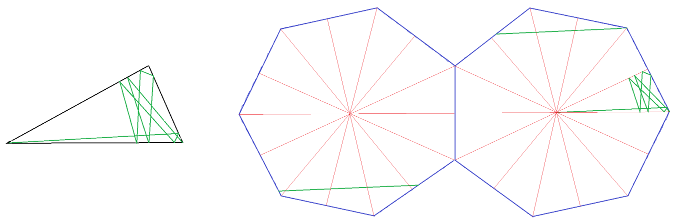

Figure 2: A vertex to vertex trajectory on the $(\frac{\pi}{2}, \frac{\pi}{7}, \frac{3\pi}{14})$-triangle in an uniquely ergodic direction.

A very natural example of Veech surface whose trace field has degree three or more over $\mathbb{Q}$ is given by the family of double regular $n-$gons, for $n \geq 7$ odd (see Figure 1). In this family of surfaces, the central points seem to be natural candidates for connection points. We show that this is false:

Theorem 2. *[Bou22, Bou25c] The central points of the double regular $n-$gons are not connection points.*

In fact, our result is more general and we also provide a large family of points that are not connection points on the double regular $n-$gon for $n \geq 7$ odd, with $n \neq 9$. We conjecture that for $n = 7$, this family is exhaustive, that is all the remaining points are connection points, see Conjecture 3.

Hecke groups and continued fractions. In fact, the Veech group of the double regular n-gons ($n \geq 3$ odd) is conjugated to the Hecke group of level n, which is a very well studied group that we define below. In particular, periodic directions on the double regular n-gon correspond to parabolic limit points on the Hecke group of level n.

The Hecke group of level $n \geq 3$ is the subgroup $\mathcal{H}_n$ of $SL_2(\mathbb{R})$ generated by $\begin{pmatrix} 1 & 2\cos\frac{\pi}{n} \\ 0 & 1 \end{pmatrix}$ and $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$. These groups were introduced and studied by E. Hecke in the 1920s, and D. Rosen [Ros54] studied its parabolic limit set. He notably demonstrated that the parabolic directions are exactly those with a finite continued fraction expansion for a variant of the continued fraction algorithm now bearing his name. However, this criterion is not entirely satisfactory as it does not allow to characterize algebraically periodic points, and it does not even allow for a finite-time test to determine whether an element represents a cusp or not. In the cases[4] $n = 3, 4, 5, 6, 8, 10, 12$, the work of several authors (especially Rosen [Ros54] and Leutbecher [Leu67]) resulted in an algebraic characterization of the set of cusps, but the problem remains largely open for all other values of n.

Question 3. *Characterize algebraically the parabolic limit set of $\mathcal{H}_n$ for odd $n \geq 7$.*

In truth, solving this question is very ambitious and constitutes rather a long-term problem over many years. For now, understanding the case $n = 7$ and getting more insight into the case $n = 9$ would already be remarkable. We actually believe that there are connection points for $n = 7$, while it is not the case for $n \geq 9$. Moreover, for $n = 7$ we have a precise conjecture on the characterization of connection points on the double heptagon, see [Bou25c, Conjecture 3.7]:

Conjecture 3. *A point P is a connection point on the double regular heptagon if its* normalized *coordinates are of the form $\frac{1}{2k}(x, y)$ where $k \geq 1$ is an integer and $(x, y) \in \mathbb{Z}[\lambda_7] \times \mathbb{Z}[\lambda_7]$ are coprime and reduces modulo two to the (finite) orbit of and element $(z, 0)$, for $z \in \mathbb{Z}[\lambda_7]/2\mathbb{Z}[\lambda_7]$.*

The idea of reducing elements of $\mathbb{Q}[2\cos\frac{\pi}{n}]$ modulo two comes from an argument by Borho and Rosenberger [Bor73, BR73] and applies for $n \geq 7$ odd, with the exception of $n = 9$ (see also [HMTY08], in English). Informally, we conjecture that the modulo two obstruction is the only obstruction for an element of $\mathbb{Q}[2\cos\frac{\pi}{7}]$ to represent the cusp of $\mathbb{H}/H_7$. For $n \geq 9$, this no longer seems to be the case, and the nature of the obstructions for a direction to represent the cusp of $\mathbb{H}/\mathcal{H}_n$ is completely mysterious. Discovering a new obstruction would be particularly remarkable.

[4]These cases correspond exactly to the cases where the trace field is $\mathbb{Q}$ or quadratic over $\mathbb{Q}$.

B - Algebraic Interaction strength

We now turn to a second research axis, in which we are interested in the algebraic intersection of closed curves on a surface equipped with a Riemannian metric, and more specifically in the following question:

How many times can two closed curves of a given length intersect?

This question, which makes sense for any orientable surface X on which we can measure the lengths of closed curves (we will assume here that the surface is equipped with a Riemannian metric with conical singularities), can be quantified by defining:

$$KVol(X) := Vol(X) \cdot \sup_{\alpha,\beta} \frac{Int(\alpha, \beta)}{l(\alpha)l(\beta)}$$

where the supremum is taken over all pairs of closed curves on X. Normalizing by the volume makes this quantity invariant under scaling. Following the terminology of [Tor23], this quantity is called the *algebraic interaction strength*. The study of KVol dates back to the work of D. Massart in [Mas97], where this quantity appears (indirectly) as a comparison constant between the stable norm and the Hodge norm. The study of KVol has since been expanded by D. Massart and B. Muetzel [MM14], but many questions remained unresolved, starting with the question of the explicit calculation of KVol for simple examples of surfaces. In this context, S. Cheboui, A. Kessi, and D. Massart initiated the study of KVol on translation surfaces by calculating KVol on the Teichmüller disk of certain square-tiled-surfaces, see [CKM21]. With E. Lanneau and D. Massart [BLM24], we provide a method that applies to non-arithmetic surfaces, for which the behavior of KVol turns out to be quite different. We later generalized this method in [Bou25b], then in [BP24] (with I. Pasquinelli) as well as in a recent work [Bou25a] where we generalize this technique to instances of translation surfaces with multiple singularities. Our main result can be summarized as follows:

Theorem 4 (B.-Pasquinelli)**.** *Let X be a translation surface made from a collection of polygons $(P_i)_{i\in I}$ whose sides are identified by pairs. We assume:*

(H1) Each polygon is convex with obtuse angles.
(H2) Sides of the same polygon are not identified.

Then, for any pair of closed curves (α, β) on X, we have:

$$\frac{Int(\alpha, \beta)}{l(\alpha)l(\beta)} \leq \frac{1}{l_0^2}$$

where l_0 is the minimal length of a side of the polygons P_i.

B.1 - Bouw-Möller surfaces and their Teichmüller disks.

The above theorem applies in particular for a family of surfaces called *Bouw-Möller surfaces*, which are indexed by two integers m, n, and which are the translation surfaces whose Veech group is a triangle group. In the case of Bouw-Möller surfaces with a single singularity, we show in [BP24] that the upper bound obtained is optimal (and is realized by two systoles that intersect once), thus obtaining the exact value of KVol. However, this bound is not optimal for Bouw-Möller surfaces with several singularities and we actually compute the optimal bound for most of the remaining cases in the recent preprint [Bou25a], extending the methods of [BP24].

Theorem 5. *Let $X = S_{m,n}$ be a Bouw-Möller surface with at least two singularities (i.e. $\gcd(m, n) \neq 1$) and $m, n \geq 8$. Then, for any pair of closed curves (α, β) on X, we have:*

$$\frac{Int(\alpha, \beta)}{l(\alpha)l(\beta)} \leq \frac{1}{2l_0^2}$$

where l_0 is the minimal length of a side of the polygons P_i. Further, this inequality is optimal if and only if $\gcd(m, n) \neq n$.

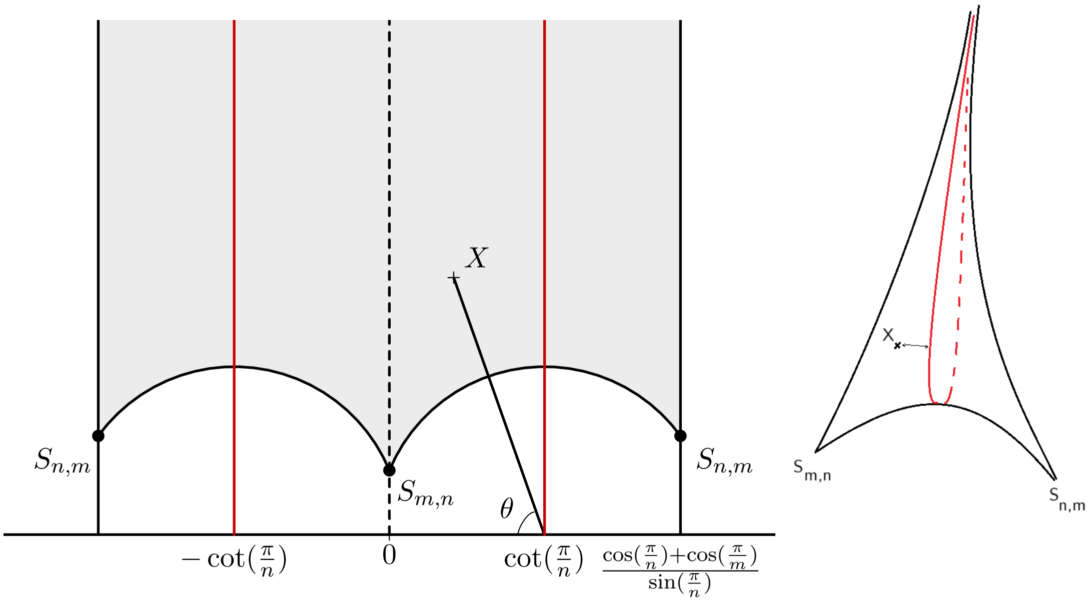

Figure 3: On the left, a fundamental domain for the Teichmüller curve $\mathbb{H}/S_{m,n}$ associated to the Bouw-Möller surface $S_{m,n}$. The geodesics $\gamma_{\infty,\pm\cot(\pi/n)}$ corresponds to the surfaces having maximal KVol, and the angle θ measures the distance between X and the geodesic $\gamma_{\infty,\cot(\pi/n)}$, namely $\cosh(\mathrm{dist}_{\mathbb{H}}(X, \gamma_{\infty,\cot(\pi/n)})) = \frac{1}{\sin\theta}$. On the right, the hyperbolic surface $\mathbb{H}/S_{m,n}$.

This result also holds for the regular $4m+2-$gon, although it is technically not a Bouw-Möller surface. If $\gcd(m,n) = n$, we believe that (apart from the case $n = 2, m = 4$) the optimal inequality requires a $1/4l_0^2$ and showing the upper bound in this case would require a much finer (and technical) analysis.

Behavior of KVol under deformation. Next, one can try to extend the result of Theorem 4 by studing the behavior of KVol under deformation. In [BLM24] and the generalizations [Bou24, BP24], we study the behaviour of KVol when the translation surfaces varies in its $SL_2(\mathbb{R})-$orbit. For this, we prove and use a sharp result of hyperbolic geometry that allows to compare the angles between two pairs of saddle connections on a translation surface. This method, first developped in [BLM24], applies particularly well to Bouw-Möller surfaces with a single singularity, for which we obtain in [BP24]:

Theorem 6. *Let $m, n \geq 2$ coprime. Given $d, d' \in \mathbb{R} \cup \{\infty\} \simeq \partial\mathbb{H}$, let $\gamma_{d,d'}$ the geodesic of the Poincaré half plane $\mathbb{H}$ whose endpoints are d and d'. Let $X = M \cdot S_{m,n}$ be a surface in the Teichmüller disk of the Bouw-Möller surface $S_{m,n}$ which is obtained from $S_{m,n}$ by applying a matrix $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbb{R})$. Then, we have:*

$$KVol(X) = K_0 \cdot \frac{1}{\cosh(\mathrm{dist}_{\mathbb{H}}(\frac{di+b}{ci+a}, \Gamma_{m,n} \cdot \gamma_{\infty,\pm\cot\frac{\pi}{n}}))}$$

where $K_0 > 0$ is an explicit constant which only depends on m and n, $\Gamma_{m,n}$ is the Veech group of $S_{m,n}$ and $\mathrm{dist}_{\mathbb{H}}$ is the hyperbolic distance in the Poincaré half plane.

A geometric interpretation of this result is given in Figure 3. We also provide in [Bou25b] a similar result in the case of the regular $4m$-gon, $m \geq 2$ (although it is a little bit more complicated to state as KVol is expressed using the distance to an infinite family of geodesics). More generally, our method allows to give a criterion to know whether KVol is bounded or not as a function on the Teichmüller disk of any Veech surface. This criterion uses the notion of separatrix diagram and makes it possible to show for example that:

Theorem 7. *KVol is bounded on the Teichmüller disk of any* algebraically primitive *Veech surface.*

The algebraically primitive Veech surfaces are those whose trace field has degree equal to their genus. It would be interesting to know whether we could use this method to estimate KVol on the Teichmüller disk of any primitive Veech surface of genus two. This would allow to answer the following question for $g = 2$:

Question 4. *Find the optimal constant $C(g) > 0$ such that for any translation surface of genus g with a single singularity, we have* $\mathrm{KVol}(X) > C(g)$.

We already know from [MM14] that $C(g) \geq 1$ and I prove in [Bou24]:

Theorem 8. $C(g) \leq g$.

We conjecture from computer experiments that this bound is optimal, that is $C(g) = g$.

Remark. *In another direction, Balacheff, Karam and Parlier [BKP18] (indirectly) show (see also Jiang-Pan [JP24] for the same result) that for hyperbolic surfaces, there exist a (universal) constant $C > 0$ such that for every hyperbolic surface X of genus g,*

$$\mathrm{KVol}(X) \geq C \frac{g}{(\log g)^2}. \tag{1}$$

B.2 - Open question and projects

We end this section with related questions concerning the interaction strength.

Supremum vs. Maximum. First, a very natural question one could ask is the following:

Question 5. *Under what condition(s) is the supremum in the definition of KVol a maximum? More specifically, is it a maximum for a Veech surface of genus at least two?*

In fact, this Question is the topic of an ongoing project with D. Massart and we are currently able to prove that the answer of the second question is *yes*.

One of the main motivations for studying KVol comes with its relation with the *stable norm* on the homology $H_1(X, \mathbb{R})$ of the surface [Mas97]. This norm, defined by Gromov [GLP81], can be pretty wild, and the unit ball is often a polytope with an infinite number of vertices [Mas97, Mon24]. In particular, when the supremum in the definition of KVol is achieved, the pairs of (normalized) homology classes that achieve the maximum are the vertices of the unit ball of the stable norm. However, if the supremum is not a maximum, this means that the unit ball has an accumulation point of vertices which is itself an extremal point.

Comparison with the systolic volume. The systolic volume $\mathrm{VolSys}(X)$ is defined as the ratio of the volume of the surface over the square of the (homological) *systole*, which is the smallest length of a non-trivial curve in homology. It has been widely studied and one can wonder how KVol compares with it. An elementary geometric argument by D. Massart and B. Muetzel [MM14] shows that for any surface equipped with a Riemannian metric X, we have $\mathrm{KVol}(X) \leq 9 \cdot \mathrm{VolSys}(X)$. Hovewer, the constant 9 is believed not to be optimal and one could try to improve it, see for example [BP24, Conjecture 1.9]. In fact, the joint result with I. Pasquinelli [BP24] discussed above allows to compare KVol and the systolic volume for a class of surfaces, and we may be able to generalize it.

Behaviour with the genus. Next, there are a lot of questions regarding the growth of KVol with the genus. This turns out to be related to the study of intersecting short curves and, as a consequence, there are similarities with the study of the systolic volume. In particular, one could ask:

(1) What is the growth of KVol with the genus for Riemannian surfaces of non-positive curvature? Without curvature assumptions?
(2) What is the growth of KVol with the genus for translation surfaces with one singularity ?
(3) Determine the infimum of KVol on the set of Riemannian surfaces of genus 2, and, if they exist, the surfaces that achieve it.

The motivation for studying these problems is to explore the properties of minimizing surfaces for KVol, and to see whether they have interesting properties. Regarding the first point, the asymptotic growth of KVol with the genus is expected to be similar to the hyperbolic case (given in Equation (1)). For translation surfaces, we expect a linear growth, which would also be coherent with the systolic volume growth [BG21, JP19]. In fact, we conjecture that the infimum of KVol for translation surfaces with a single singularity is exactly g, see [Bou24]. In fact, solving the case $g = 2$ would already be remarkable, see Question 4.

What about the geometric intersection? In the previous paragraphs, we considered the *algebraic intersection* motivated in part by the relation of KVol with the stable norm, but the previous questions can be phrased similarly for geometric intersection, as studied for example in [Tor23]. One could wonder what are the similarities and differences between the *geometric* and the *algebraic* interaction strength. For example, it should be first noticed that for any surface X where both quantities are well defined, the geometric interaction strength is always bigger (or equal) than its algebraic counterpart. However, the inequality can be realized or strict, even for translation surfaces with a single singularity. In general, we expect that many results valid for one will remain valid for the other, and may even be easier to prove in the geometric framework (for example, Question 5). See [Bou25a] for a discussion on the geometric interaction strength).

C - Automorphisms of translation surfaces

We now move to another recent research project, in collaboration with R. Gutiérrez-Romo and E. Lanneau, where we study the automorphisms of translation surfaces, which are locally given by translations. Schlage-Puchta and Weitze-Schmidthusen [SPWS17] proved that the automorphism group of a given translation surface of genus g has at most $4(g-1)$ elements and, by analogy with the automorphisms of Riemann surfaces, call the surfaces that realize this bound *Hurwitz translation surfaces*. They also prove that this bound is achieved only for certain genera (g odd or of the form $3k+1$). In the recent preprint [BGRL25] we prove:

Theorem 9. *Let $g \geq 2$. We have the equality*

$$\sup \left\{ |\mathrm{Aut}(X)| \;\middle|\; X \text{ is a genus-}g\text{ compact translation surface} \right\} = \frac{(m(g)+1)}{m(g)}(2g-2),$$

where $m(g) \in \mathbb{N} \cup \{\infty\}$ is the maximal order of the singularities of a regular origami in genus g, and it satisfies $m \mid 2(g-1)$, $3 \nmid m$, and $4 \nmid m$. Finally, the case $m(g) = \infty$ arises if and only if no genus-g regular origamis exist.

The explicit determination of the genera such that $m(g) = 1$ corresponds to the work [SPWS17]. This result relies on the study of *regular origamis*, which are regular covers of the square torus ramified at a single point. We show that these surfaces achieve the best possible number of translations in their genus (provided they have the maximal possible number of singularities among the regular origamis in a given genus). In fact, these surfaces can be constructed from a finite group generated by two elements, and we can study their existence in a given genus using group-theoretic tools. Namely, we prove:

Theorem 10. *Let g be of any of the following forms:*

- $g = p + 1$, *where $p \geq 5$ is prime;*
- $g = p^2 + 1$, *where p is prime, but is not a Sophie Germain prime;*
- $g = pq + 1$, *where $p, q \geq 5$ are distinct primes.*

Then there is not regular origami in genus g.
Further, in the case $g = p^2 + 1$ and $p \geq 5$ is a Sophie Germain prime, the only regular origamis in genus g have group $\mathbb{Z}/(2p+1)\mathbb{Z} \rtimes \mathbb{Z}/p\mathbb{Z}$.

Recall that a *Sophie Germain prime* is a prime p such that $2p + 1$ is also prime. The proof of this result is based on analyzing the existence of regular origamis whose associated abelian differentials have zeros of specific types, that is, belonging to particular *strata*. We focus on the cases $g = p+1$ and $g = pq+1$ primarily because, in these situations, the Euler characteristic $2-2g$ admits a simple prime factorization, and the problem reduces to finding groups with a cyclic subgroup of prime or twice-prime index. One may seek to generalize these results to the case where the genus has a specific prime factor decomposition, e.g. the product of three disctinct primes or $p^\alpha + 1$ where p is a prime. However, these cases seem to be more complicated.

Let us point out that the full automorphism group (and some of its subgroups) of a Riemann surface of genus $g = p+1$ and $g = p^2+1$ has been considered before in the literature [BJ05, IJRC21, CRC21]. While the group-theoretic setting differs, the simple factorization of the Euler characteristic similarly facilitates the analysis.

Finally, let us also mention that there are a lot of interesting questions related to the surfaces studied in this project, such as:

Question 6. *Does there exist a origami of genus g whose cylinder decompositions (in every periodic direction) all have g cylinders?*

D - Other projects

D.1 - Khinchin's Constant for Interval Exchange Transformations (IETs), and Zorich measure

In this project, in collaboration with R. Gutierrez-Romo, E. Lanneau, V. Maturana, and A. Pardo, we focus on the notion of *Khinchin's constant*. The Khinchin's constant $K \simeq 2.68545\ldots$ is the almost-sure limit of the geometric mean of the integers appearing in the continued fraction expansion of a generic real number $x = [a_1, a_2, \ldots] \in [0,1]$. It also corresponds to (the exponential of) the ergodic average of the function $f(x) = \log(a_1)$, for the Gauss map $g(x) = 1/x - \lfloor 1/x \rfloor$. In particular, since continued fractions are in correspondence with rotations on the circle, and thus with interval exchanges on $d = 2$ intervals, one can seek to generalize Khinchin's constant to the case of exchanges of $d \geq 3$ intervals. This generalization is made possible by the existence of a finite measure on the set of Interval exchange transformations on $d \geq 2$ intervals, as demonstrated by Zorich [Zor96], and in this context the continued fraction expansion is replaced by *Rauzy induction* (see for example [Yoc05]): to each *Rauzy diagram* $\mathcal{R}$ (see Figure 4) as associated a Khinchin constant $K_{\mathcal{R}}$ corresponding to the geometric average of consecutive *top* and *bottom* operations obtained by the Rauzy induction on a generic permutation.

This measure, now called the *Zorich measure*, has been the subject of many questions over the past twenty years, and is notably related to the Masur-Veech measure associated with translation surfaces[5]. Numerous ideas have been developped with the aim to understand this measure, see for example [AE01, AE03, DGZZ21, DGZZ22]. However, these approaches use the paradigm of translation surfaces (counting square-tiled surfaces, Siegel-Veech constants), and they do not allow access to the measures associated with each permutation, for example.

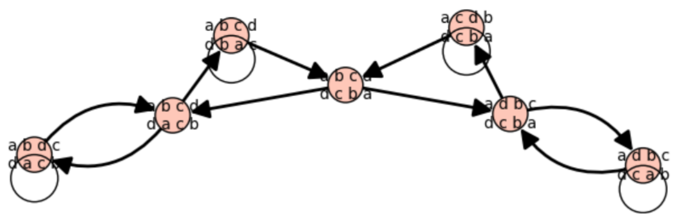

Figure 4: The Rauzy diagram associated with the symmetric permutation on four intervals.

D.2 - Abelian Complexity of Words coming from IETs.

I am also interested in the abelian complexity of words arising from interval exchanges. I learned about this topic during a semester working group with M Andrieu R. Gutierrez-Romo, V. Maturana, and A. Pardo. First, the complexity of an infinite word $\omega \in \{1, \ldots, d\}^{\mathbb{N}}$ is the function $p : \mathbb{N} \to \mathbb{N}$ which, to an integer n, associates

[5] Translation surfaces are naturally obtained as suspensions of IETs.

the number of distinct words of size n found in ω, in other words, the cardinality of the language of size n associated with ω, defined by $\mathcal{L}_\omega(n) = \{u|u \text{ is a factor of size } n \text{ of } \omega\}$. Now, the abelian complexity $\rho : \mathbb{N} \to \mathbb{N}$ is the function obtained by counting the words of size n up to abelian equivalence: two words are considered identical when the number of occurrences of each letter is the same, i.e., they can be transformed into each other by changing the order of the factors.

The notion of abelian complexity was introduced by G. Richomme, K. Saari, and L.Q. Zamboni [RSZ09], and its main interest is that, similarly to classical complexity, it characterizes both purely periodic words: these are words ω such that there exists an $n \geq 1$ with $\rho_\omega(n) = 1$, and Sturmian words, which are words ω such that $\rho_\omega(n) = 2$ for all $n \geq 1$. In this context, one may be interested in the abelian complexity of certain families of words. One such class is given by the words arising from minimal interval exchanges, where the word associated with a real number $x \in [0, 1]$ corresponds to the coding of the intervals encountered along its trajectory. Minimality implies that the language of each word arising from a given interval (for some $x \in [0, 1]$) is the same, and thus the same holds for complexity and abelian complexity.

In this context, the first question one can ask is whether the abelian complexity is bounded or not. This question is related to the notion of *balance* of the word as well as to *deviations* from the ergodic averages, which are questions that have been, and continue to be, widely studied, see for instance [Zor97, Ada03, MMY10, BBH14, TU22]. In particular, the foundational work of Zorich [Zor97] relates deviations to Lyapunov exponents associated with the Kontsevich-Zorich cocycle.

D.3 - Other interests

Apart from the questions presented here, I am also eager to work on other projects. I participated in several conferences and workshops of the ANR Adyct on **spectral/microlocal analysis on surfaces**, and I am also interested in **Teichmüller theory** and **pseudo-Anosov diffeomorphisms**. For example, in the last meeting of the ANR Adyct, in July 2025, I presented a paper of Eskin and Mirzakhani [EM11] on the count of closed Teichmüller geodesics in moduli spaces. The Teichmüller geodesic flow comes from the $SL_2(\mathbb{R})-$action on (half-) translation surfaces, and closed geodesics corresponds to (conjugacy classes) of pseudo-Anosov diffeomorphisms. I am still interested by that topic and would be happy to discuss it! Finally, I enjoy working with Sagemath to test my questions on simple examples and develop intuition.

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